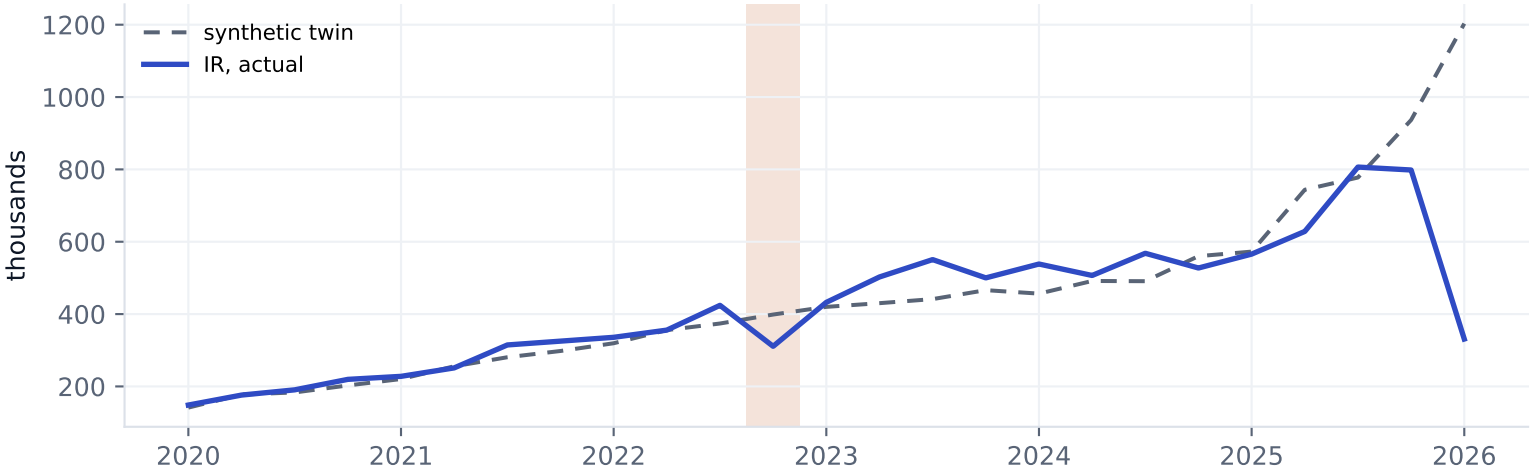
Iran, 2022

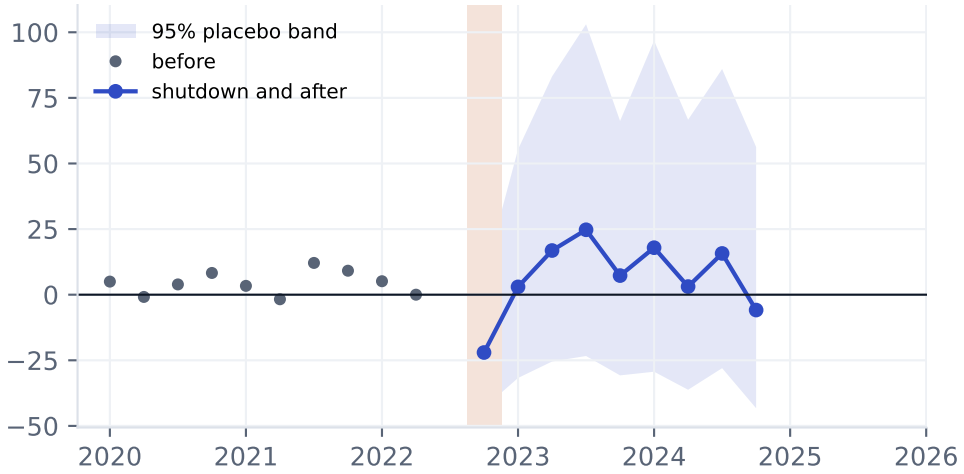
Change in git pushes attributed to IR in 2022Q4 vs a synthetic twin (106 donor economies, 10 pre quarters)

Method	vs twin	95% interval	p normal	p perm
Synthetic DiD	-22.0%	-43% to +7%	0.122	0.075
Synthetic control	-23.8%	-47% to +10%	0.144	0.112
Augmented synthetic control	-23.0%	-43% to +5%	0.097	0.103

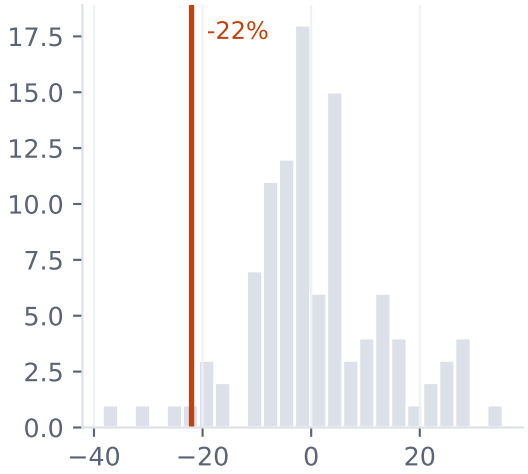
Git pushes per quarter, attributed to IR



Gap to the twin (%)



Same design run on every donor (%)



- Conformal test. p = 0.091. With 11 quarters the smallest possible p is 0.091, so this test can't give an interval here.
- Backdating. Fitting without 2022Q1 and 2022Q2, the twin misses those quarters by -18%, -29%. Outside the placebo band, so the pre trend is shaky.
- Breakdown. The interval already includes zero, so there is nothing to break.
- Leave one donor out. Dropping each of the 28 donors with weight over 1% keeps the estimate between -22.9% and -20.5%.
- Holm adjusted p across the nine registry events is 0.664. Does not pass every check in the analysis plan, so I don't call it a clear drop.
- What happened. Mobile networks cut 4pm to midnight from 21 Sep to 3 Oct 2022 and again on 8, 12 and 15 Oct, regional outages, app stores and messaging apps blocked. Fixed line mostly stayed up. Source https://ooni.org/post/2022-iran-technical-multistakeholder-report/

GitHub activity placed in the country by IP address, not national output. VPN users get placed where the VPN exits.